

Build a guaranteed code breaker

Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

1. Play four places.

Try a hidden code. Then build a method that always identifies it in at most **four scored tests**. You may name the answer without paying for a final test.

Test	My code	Score	Still possible
1			
2			
3			
4			

2. Explain a baseline method.

Test RRRR. Then separately change place 1, place 2, and place 3 to B, keeping every other place R. Explain how the fourth place is recovered.

What each score change tells me, and why the method covers every secret:

A real optimization question: Four tests are enough. Could a cleverer, adaptive method always manage three? On the next page, play the role of a difficult secret maker.

Build first. Explain by pointing, talking, or drawing.

The secret maker fights back

Name: _____

Date: _____

Two colors: R and B. Repeats allowed. A test earns one point for each color in the correct position. You hear **only the total**, not which positions match.

3. Make the first answer difficult.

After RRRR, answer 2. These are the six possible secrets. Group them by their score for each proposed second test.

BBRR BRBR BRRB
RBBR RBRB RRBB

Second test	Score groups	Largest group	Its size
BBRR			
BBRR			
BBBR			
BBBB			

4. Only one test remains.

For BRRR, let the answer leave three candidates. Can one last test separate those three? For BBRR, let the answer leave four candidates. Can one last test separate those four?

Try final tests; explain why each difficult branch defeats them.

Three-secret hint: Use BBRR, BRBR, BRRB. Which position is fixed? What is the only feature that changes? A test asks whether the moving B falls in one of its B positions. How many scores can that give?

Four-secret hint: All six original candidates contain exactly two Bs. How many scores can one test produce on them? Is that enough to separate four candidates?

Build first. Explain by pointing, talking, or drawing.

Turn the experiment into a proof

Name: _____

Date: _____

5. Cover every possible second test.

Second tests have 0, 1, 2, 3, or 4 Bs. Changing position names does not change the problem. Swapping every R and B in a test replaces its score by $4 - \text{score}$. Use those facts to explain why the cases on page 2 cover everything.

Why one or three Bs leaves a hard three-secret branch:

Why two Bs leaves a hard four-secret branch:

6. What about a different first test?

Rename the two colors separately at selected positions, in both every secret and every test. Can you turn any first test into RRRR without changing which places match?

My conclusion: the smallest guaranteed number of tests, and why:

A short method proves an **upper bound**. A difficult branch that defeats every shorter method proves a **lower bound**. When they meet, we know the optimum.

Go further: Must five places need five tests? The extra page contains a surprise.

Build first. Explain by pointing, talking, or drawing.